

Divorce-induced health decline and the impact on mortality risk*

Kar Man Tan[†]

Pavel Grigoriev[‡]

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Abstract

Divorce is linked to elevated mortality. However, the causal mechanisms underlying this relationship are complex and remain insufficiently identified. Health is one such mechanism and is the focus of this paper. Using linked administrative data from the German Pension Insurance Fund (DRV), we examine whether divorce increases the risk of an initial musculoskeletal rehabilitation, and how much of the divorce-mortality gap runs through this channel. Rehabilitation is administered by the DRV and is granted only once a musculoskeletal condition has become chronic and threatens the ability to work. Our outcome measures the crossing of an institutional threshold rather than the onset of disease. We build annual risk sets of individuals experiencing a first (observable) divorce and individuals who have not yet divorced in that year and then use propensity score matching to generate comparable matches. **Results: Divorce effect on cumulative incidence of musculoskeletal rehab, divorce effect on mortality, results from decomposition**

Keywords: divorce, health, rehabilitation, mortality

JEL Classification: H75, I12, J12

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[†]Corresponding author: karman.tan@bib.bund.de. Federal Institute for Population Research (BiB), Friedrich-Ebert-Allee 4, 65185 Wiesbaden, Germany

[‡]Federal Institute for Population Research (BiB)

1 Introduction

The link between divorce and mortality has been widely discussed in social epidemiological studies. Register-based studies have used Finnish data (Metsä-Simola and Martikainen, 2013), Norwegian data (Berntsen and Kravdal, 2012) and English/Welsh data (Grundy and Tomassini, 2010). Until recently, Germany has been a difficult setting for this question. Brockmann and Klein (2004) is the only study to the best of our knowledge which explicitly analyzes the link between divorce and mortality although the dataset used in the study is the SOEP¹ which is not an administrative dataset. Since 2011, it is possible to link different administrative datasets from the German Pension Insurance Fund (DRV) using a unique identifier, allowing us to access full income biographies, divorce information², rehabilitation history and death. The availability of this dataset led to the conception of the project *Divorce and Diagnosis (DiDi): Register-based Analysis on the Role of Divorce in Health and Employment Dynamics*³, which constitutes our study.

Our study focuses on the interaction between divorce, health and mortality. While there can be a direct link between divorce and mortality if divorce results in suicide (Kposowa, 2000; Stevenson and Wolfers, 2006) or homicide, causal evidence remains scarce when the link is indirect. Hence, we chose health as it is one possible indirect link tying together divorce and mortality. There can, however, still be threats to identification as health is not only a consequence but also a cause of divorce (Torvik et al., 2015). Furthermore, the individuals who choose to divorce may not necessarily be random. Many of the register-based studies make use of hazard models to estimate excess mortality risk after divorce and controls for sociodemographic and socioeconomic variables. For Germany, Brockmann and Klein (2004) account for health by using the self-reported health measure from the German SOEP and control for variables associated with marital status to find that divorced men have 51% higher mortality than married men while there is no significant effect for women. Leopold (2018) presents a very careful multi-outcome treatment of the gendered consequences of divorce over time, but it is not designed to control for the health that occurs before from the health consequences which follow the divorce. In our paper, we control for pre-existing health trends and other factors which occur before the divorce by using propensity score matching to answer the following questions

RQ1: Does divorce raise the risk of musculoskeletal rehabilitation⁴ among divorced and non-divorced individuals?

RQ2: Is some of the excess mortality caused by divorce occurring because divorce caused additional musculoskeletal deterioration?

For context, the DRV only grants rehabilitation once the musculoskeletal condition is chronic to the point that it threatens working ability (Deutsche Rentenversicherung Bund, 2005). We do not argue that divorce causes musculoskeletal disease but rather whether divorce moves individuals who carry a latent vulnerability across the legal

¹German Socioeconomic Panel

²This information is available due to the law of splitting pension points.

³This project is a collaboration between the Federal Institute for Population Research, Hertie School, Charité - Universitätsmedizin Berlin and the DRV.

⁴Musculoskeletal diseases are the most common cause of lost work time in Germany.

threshold where rehabilitation becomes necessary. Please see Section 2.3 for a more detailed discussion.

Results... maybe policy implications?

Section 2 provides a literature review, Section 3 introduces the data and methodology, Section 4 presents the results, Section 5 discusses the implications and Section 6 concludes.

2 Literature review

2.1 Link between divorce and mortality

The link between divorce and mortality risk is widely documented in sociology and epidemiology. In a meta-analysis, Shor et al. (2012) run meta-regressions on 625 mortality risk estimates over 104 studies and find that the mean hazard ratio was 1.30 for divorced individuals. The hazard ratio is slightly higher for men at 1.37 and lower for women at 1.22, although the difference between genders decrease with age. However, heterogeneity on the effect on death exists depending on the time since divorce. Metsä-Simola and Martikainen (2013) finds that excess mortality among Finnish men, tied to accidental, violent or alcohol-related deaths, is highest immediately after divorce and declines over time. On the other hand, Berntsen and Kravdal (2012) finds that excess mortality among Norwegian men increases with time since divorce and remarriage can change the magnitude of the effect. In Germany, the evidence is more limited. 2004 (2004) use event-history models which account for marital transitions and duration in each marital status. They find evidence of selection into marriage and that mortality risk is lower when the marriage duration is longer.

Most of the previous studies establish a robust association, however, causal evidence remains scarce as divorce is not random. Sbarra and Nietert (2009) follow 1376 individuals in the Charleston Heart Study for forty years and show that the definition of the exposure determines the size of the effect. Individuals who are continuously separated or divorced have a higher hazard ratio (1.66) compared to individuals who have ever been separated or divorced (0.98). Wiborg and Gugushvili (2025) initially find raised hazard ratios for divorced men (1.59) and women (1.47) using Norwegian data. After using twin-pairs to account for genetic and early developmental and environmental factors, the relationship between divorce and mortality disappears.

2.2 Divorce and health

The effect of divorce on health is difficult to isolate due to the possibility of reverse causality. Joung et al. (1998), using Dutch data, find that married individuals who had more subject health complaints were 1.5 times more likely to divorce, while individuals who reported multiple chronic conditions were twice as likely to divorce. Torvik et al. (2015) corroborates Joung et al. (1998)'s findings with Norwegian data. Poor health behaviors including smoking, drinking, and lack of exercise in addition to mental distress predict subsequent divorce. Metsä-Simola and Martikainen (2013), with Finnish data, find that psychotropic medication use among individuals who divorce is

already elevated five years before hand and peaks at around six to nine months just before divorce. Wade and Pevalin (2004) find worse mental health among those divorcing or divorced than among the married using British data. Generally, poor health tends to precede divorce.

To deal with health selection, Kohn and Averett (2014) estimates a dynamic model that accounts for the persistence of health and how it interacts with relationship status using 18 waves of the British Household Panel Survey. The result is that divorce has a marginally significant effect on the health of men younger than 45. Sbarra et al. (2014) also finds only a marginal effect of divorce on health. After matching individuals on divorce probability, Sbarra et al. (2014) finds that only individuals who were already depressed before divorce experienced major depressive episodes following divorce. With respect to mental health, Björkenstam et al. (2013) find that divorced individuals have thrice the risk of psychiatric inpatient care compared with married individuals in Sweden. Meulman et al. (2023) find similar results in the Netherlands where divorced individuals had higher health expenditures dedicated to mental health care. In terms of physical health, the evidence is heterogeneous. Zulkarnain and Korenman (2019) find that women who divorce at older ages tend to have more physical changes than men. Leopold (2018), on the other hand, find that men are more vulnerable in the short run in terms of well-being although the gender differences narrow in the medium run. Pellón-Elexpuru et al. (2024), in a meta-analysis across 94 studies find that divorced individuals report worse self-rated physical conditions with higher complaints of musculoskeletal, cardiovascular and cerebrovascular conditions although the effect is not significant.

2.3 Divorce and musculoskeletal disease

Evidence on the pathway from divorce to musculoskeletal impairment is rather limited and ambiguous. However, we chose to focus on musculoskeletal diseases because they are a major source of disability and work interruption in Germany⁵. The DRV

⁵For an individual covered by statutory health insurance, musculoskeletal pain first enters the health insurance system through sickness absence rather than through the DRV. If pain makes an employee unable to work, the individual must immediately inform the employer of the incapacity and the expected duration. If the incapacity lasts longer than three calendar days, a medical certificate needs to be acquired from the individual's general practitioner, although the employer may require certification from the first day. The employer continues to pay the normal salary for up to six weeks for incapacity caused by the same illness (Bundesrepublik Deutschland, 1994). After the six weeks, if the illness continues, the statutory health insurer steps in and pays *Krankengeld*, which is usually about 70% of covered gross income up to a maximum of 90% of net income. For the same illness, this benefit is limited to 78 weeks within a 3-year period, counted from the first day of incapacity, meaning that the time of illness paid by the employer is included in the 78 weeks (Bundesrepublik Deutschland, 1988). Medical rehabilitation facilitated by the DRV requires a separate application and assessment of whether earning capacity is substantially endangered or already reduced, if rehabilitation is expected to prevent or improve that reduction (Bundesrepublik Deutschland, 1989; Deutsche Rentenversicherung, n.d.). This application is usually independent of the six weeks and *Krankengeld* although the health insurer can allow the individual ten weeks to submit a rehabilitation application indicating that earning capacity is threatened (Bundesrepublik Deutschland, 1988). For musculoskeletal conditions, the DRV assessment focuses on documented functional limitations and their consequences for work and social participation. Chronic pain, psychosocial circumstances surrounding incapacity, comorbidity, and active cooperation are considerations in this assessment rather than four cumulative eligibility conditions. Isolated pain is insufficient to establish rehabilitation need for certain hip, knee, and shoulder conditions (Deutsche Rentenversicherung

Reha-Atlas (Deutsche Rentenversicherung Bund, 2024) reports of all completed medical rehabilitations, orthopedic conditions are the most common, accounting for 35.6% of rehabilitations among women and 35.5% among men. Brenscheidt et al. (2026) reports that musculoskeletal diseases account for 21% of certified incapacity days in 2024, which is the largest diagnostic group.

Sjögren et al. (2009), using Danish data, find that chronic pain is more common among individuals who are divorced, separated or widowed, where musculoskeletal diseases accounted for 66.8% of the cause of chronic pain. However, Choi and Wagner (2026), using the Health and Retirement Survey find that individuals who transition from marriage to divorce tend to have a lower probability of high-impact pain. On the other hand, Cimas et al. (2026) using SHARE⁶ data, find that there are gender differences. While divorced women have a higher incidence of chronic musculoskeletal pain, men have a lower prevalence. Meanwhile, Andorsen et al. (2017) finds that there is no significant effect between divorce and musculoskeletal complaints using Norwegian data.

While the impact of divorce on musculoskeletal diseases is ambiguous, the evidence on the relationship between social support and the consequences of musculoskeletal diseases is more homogeneous. Prang et al. (2015) finds that individuals who have support from family friends and neighbors tend to have less persistent pain associated with musculoskeletal diseases, while widowed, separated or divorced individuals, in the absence of a network, tend to have a higher probability of persistent pain. Woods (2005) corroborates this and finds that a lack of support is associated with higher musculoskeletal morbidity, absence from work and subsequently, failure to return to work. Reese et al. (2010) finds that the quality of marriages matter, i.e. individuals with arthritis in problematic marriages tend to have more pain complaints and disability compared to individuals in well-adjusted marriages. This could signify that a decline in the relationship quality prior to divorce leads to poorer support, hence more complaints of pain. Couch et al. (2015) and Tamborini et al. (2016) find that divorce is associated with subsequent health threatening the ability to work, especially among individuals who do not remarry, although they do not differentiate by types of diseases.

2.4 Musculoskeletal disease and death

Musculoskeletal conditions are rarely direct causes of death in the way that cardiovascular diseases are. They often operate indirectly, for example through inflammation affecting the cardiovascular disease risk. Rheumatoid arthritis and chronic inflammatory activity can contribute to oxidative stress, dysfunction of blood vessels, arterial build-up and other cardiovascular risks (England et al., 2018). Cardiovascular mortality among people with rheumatoid arthritis is about 60% higher than the general population (Meulman et al., 2009). For non-inflammatory conditions, musculoskeletal conditions causes pain and restricted movement, which then makes

Bund, 2005). Approved medical rehabilitation is normally provided on an inpatient or full-day outpatient basis and usually lasts approximately three weeks. The employer continues wages during rehabilitation if entitlement remains, otherwise, the DRV may pay "*Übergangsgeld*", normally 68% of previous net earnings or 75% for an insured person with a qualifying child, while entitlement to *Krankengeld* is suspended (Deutsche Rentenversicherung; Bundesrepublik Deutschland, n.d. 1988).

⁶Survey of Health, Aging and Retirement in Europe

people avoid physical activity. This reduces cardiovascular health and fitness, which then potentially increases the risk of comorbidities. Nüesch et al. (2011) finds that English patients with knee or hip osteoarthritis tend to have higher all-cause and cardiovascular mortality. Hawker et al. (2014) finds a similar result with Canadian patients.

Jansson, Mittendorfer-Rutz, and Alexanderson (2012) is more related to our study. Using Swedish register data, the authors compared individuals who were absent from work due to musculoskeletal diseases with people who were absent from work due to other diseases and people who were not absent from work. They find that among three categories of musculoskeletal diseases, the all-cause morbidity was 1.3 to 1.5 times higher. Given that musculoskeletal diseases account for the highest share of rehabilitations facilitated by the DRV, it is important that rehabilitation can encourage return to work and by extension, reduce the all-cause mortality attributed to musculoskeletal diseases. Bethge et al. (2019) finds that work-related medical rehabilitation tends to increase the return to work rate by 5.8 percentage points and shortens the median time to return by 9.46 days compared to conventional medical rehabilitation. In our study, the first completed DRV-facilitated musculoskeletal rehabilitation should be interpreted as an indicator of serious, work-relevant impairment and also receipt of potentially beneficial treatment.

3 Data and methodology

3.1 Data

The Research Data Center of the DRV allows linkages of different datasets starting from the reporting year 2011 through a unique identifier (Forschungsdatenzentrum Rentenversicherung, 2025). The datasets we use are AKVS⁷, LTVS⁸, RTBN⁹, RTWF¹⁰, RSD¹¹ and VA¹². The time period covered by the data is 2011 to 2023.

The AKVS, LTVS and RTBN are merged by the DRV to form the BASE dataset. The AKVS dataset contains individuals who are still actively contributing to the pension fund hence it generally represents individuals who are employed. The LTVS dataset contains individuals who have paused pension fund contributions but have not yet retired (nor drawn from their pension fund). It may include individuals who are unemployed, undergoing further education, on parental leave, etc. The RTBN dataset contains individuals who are drawing from their pension fund, i.e. retired individuals. The merged dataset, BASE, allows us to map out the full life-course coverage of individuals. If we only considered only RTBN, the sample is biased towards individuals who have contributed and lived long enough to draw a pension, ignoring those individuals who have undergone rehabilitation or died before drawing a pension. Considering only AKVS ignores rehabilitation efforts and gray divorces. To round up the dataset, it is also important to consider LTVS which details what happens between

⁷Active Insured Persons

⁸Latent Insured Persons

⁹Insured Persons' Pension Portfolio

¹⁰Termination of Pensions due to Death

¹¹Completed Rehabilitation in the Insurance History

¹²Pension Equalization Statistics

employment spells and retirement.

We then supplement the merged dataset with the RTWF, RSD and the VA. The RTWF details pension terminations due to deaths. If the death information from the merged dataset is inconsistent with the information in the RTWF, the individuals (cases) are dropped {about 127 individuals are dropped}. The RSD dataset details the completed rehabilitations and include information such as the broad diagnosis group, the type of rehabilitation benefits applied for, start and end date of rehabilitation. The RSD dataset also includes individuals with withdrawn, transferred or rejected applications. We only include individuals whose applications were successful and who completed the rehabilitation spell. Furthermore, about 5% of individuals undergo three or more rehabilitation spells per year. We also drop these individuals. We then supplement the merged dataset by including the additional rehabilitation information for the remaining individuals in the RSD dataset. The VA dataset includes information related to divorce and only contains divorced individuals. From the VA, we can obtain the date of divorce, pension supplements and deductions which result from divorce.

3.2 Matching

Since divorce is not randomly assigned, divorced and non-divorced individuals may differ systematically before divorce occurs. We therefore use 1:1 matching without replacement to construct a comparison group of individuals who are similar with respect to observed pre-treatment characteristics. The causal interpretation relies on conditional exchangeability (Rosenbaum and Rubin, 1983). This means that, conditional on the observed characteristics used in matching, individuals who divorce and comparable individuals who do not divorce are assumed to have had similar potential health trajectories in the absence of divorce. In other words, it is assumed that there are no remaining unobserved factors that jointly affect the probability of divorce and the subsequent health outcomes. However, the administrative data may not fully capture factors such as relationship conflict or social support.

We define treatment as the first observed divorce from the earliest observed marriage spell. We limit the treatment cohorts to divorces observed between 2012 and 2018 to ensure that each cohort can be observed for at least five years after divorce. The year of divorce is fixed as t_0 . For each treatment year, we form a separate within-year risk set. For treated individuals whose first observed divorce occurs in t_0 , eligible controls are individuals who have not yet experienced a divorce by that year. Controls may divorce later. We do not restrict the control group to individuals who never divorce because this would define control eligibility using information that occurs after t_0 . This means that it is possible that an individual who was a control before their own divorce can be counted as treated afterwards. Individuals must be alive at t_0 and must have an observed record in $t_0 - 1$. We exclude individuals with a musculoskeletal rehabilitation up to and including t_0 . They may previously have received rehabilitation for other conditions. The resulting sample therefore consists of individuals who are at risk of their first observed musculoskeletal rehabilitation after t_0 .

We require treated and control individuals to be matched exactly on sex, five-year age band and whether any rehabilitation has occurred by $t_0 - 1$. Then, we estimate a logistic propensity-score model separately for each treatment year. The propensity score

represents the estimated probability of divorce in t_0 , conditional on observed characteristics measured before divorce. The characteristics we include are age, cumulative rehabilitation history, mental-health rehabilitation history, years since the most recent rehabilitation, pension contribution periods,¹³ earnings, pension income, unsuccessful rehabilitation applications, federal state, broad occupational group, income-source status and marital status at $t_0 - 1$.

We treat missingness in earnings, pension income and contribution periods as potentially informative. In particular, observed earnings with missing pension income generally indicate active employment, whereas observed pension income with missing earnings generally indicates pension receipt. We retain the observed earnings and pension values and include a categorical measure describing the joint observability of the two income sources. We also include separate indicators for missing contribution-period information. This allows the matching model to use both the recorded values and the information contained in their missingness patterns.

Within each treatment year and exact-match group, we use optimal matching. The matching procedure first maximizes the number of treated individuals who can be matched within the caliper and then chooses the combination of pairs that minimizes the total propensity-score distance. We limit matches to a maximum absolute propensity-score difference of 0.01. Matching is conducted without replacement, meaning that each treated individual and each control can appear in only one matched pair. To accommodate the large annual control pools, the propensity-score model is fitted using all treated individuals and, where necessary, a reproducibly selected sample of controls, with the total estimation sample limited to 500,000 observations. The resulting propensity scores are corrected for the control-sampling fraction, and the fitted model is then used to calculate propensity scores for every eligible treated individual and control. The computational limit therefore affects how the propensity model is estimated but does not remove controls from the matching pool.

We assess covariate balance before and after matching using standardized mean differences. Balance is assessed separately for continuous variables, binary variables, each level of categorical variables and indicators of missingness. We present these results in Figure blablabla. An absolute standardized mean difference below 0.10 is treated as evidence of adequate balance on the corresponding observed characteristic. Where residual imbalance above this threshold remains after matching, the affected variables are additionally included in the outcome model. This adjustment is intended to address limited remaining imbalance and is not used as a substitute for unsuccessful matching.

Using the matched sample, we estimate the average treatment effect among the matched treated individuals. We examine first entry into musculoskeletal rehabilitation, time to first musculoskeletal rehabilitation, rehabilitation duration and all-cause mortality after divorce. Musculoskeletal rehabilitation is examined over a five-year follow-up period, with death treated as a competing event. Mortality is followed for up to ten years where permitted by the end of the observation period in 2023. The amount of available mortality follow-up therefore varies across treatment cohorts.

¹³Periods during which pension contributions were recorded.

3.3 Identification

For identification, we focus on the average treatment effect among the matched treated individuals.

3.3.1 RQ1: Musculoskeletal rehabilitation

We follow each matched individual from t_0+1 to t_0+5 and estimate a discrete-time hazard model:

$$\begin{aligned}\lambda_{i\tau}^{\text{rehab}} &= \Pr(R_{i\tau} = 1 | \text{no musculoskeletal rehab before } \tau, \text{ alive at } \tau) \\ &= \Lambda_{\text{rehab}}(\alpha_\tau + \beta D_i + \mathbf{x}_i' \gamma)\end{aligned}\tag{1}$$

where $R_{i\tau}$ is a dummy indicating individual i 's first entry into musculoskeletal rehabilitation in time in follow-up year $t_0 + \tau$ where $\tau \in \{1, \dots, 5\}$, α_τ are fixed effects, D_i is a dummy indicating divorce at t_0 and $\mathbf{x}_i$ are other variables measured at $t_0 - 1$ that we add depending on the specification. We estimate two specifications for Equation 1. In the first specification, we omit $\mathbf{x}_i$ and rely only on matching to deal with selection bias. In the second specification, we include variables whose absolute standardized mean difference after matching exceed 0.10 into $\mathbf{x}_i$. Standard errors are clustered at the level of the matched pair.

The hazard odds ratio from Equation 1 is difficult to interpret on its own as it does not show how frequently rehabilitation occurs, hence we also estimate the cumulative incidence of a first musculoskeletal rehabilitation over the 5-year follow up period using a discrete Aalen-Johansen estimator¹⁴ (Aalen and Johansen, 1978). We report four values at the the 5-year horizon, 1) cumulative rehabilitation risk among treated individuals, 2) cumulative rehabilitation risk among controls, 3) risk difference per 1000 individuals and 4) the corresponding risk ratio. Confidence intervals are obtained by resampling whole matched pairs.

The biggest threat to the conditional exchangeability assumption is unobserved pre-divorce trends, which available variables may not fully capture, especially by only observing the year prior to the divorce. To address this, we first conducted an event study from $t_0 - 3$ to $t_0 + 5$ by taking the difference in values of unsuccessful rehabilitation applications, earnings, pension income, missingness of pension income and prior rehabilitation activity between the treated and control in each matched pair

¹⁴This estimator recognizes that individuals who die can no longer enter rehabilitation. For example, suppose there are 100 people where 10 enter rehabilitation in $t_0 + 1$, 20 die before rehabilitation and 70 are still alive and have not entered rehabilitation. Suppose in $t_0 + 2$, 7 people out of the remaining 70 enter rehabilitation. If we treat death as censoring rather than as a competing event, the proportion surviving up to $t_0 + 1$ is $1 - \frac{10}{100} = 0.9$, and up to $t_0 + 2$ is $1 - \frac{7}{70} = 0.9$, $\therefore$ the cumulative rehabilitation probability is $1 - (0.9 \times 0.9) = 0.19$, which is an overestimation. With the Aalen-Johansen estimator, we consider the proportion which actually entered rehabilitation by each follow-up year before dying. In $t_0 + 1$, the proportion entering rehabilitation is 0.1, however, after $t_0 + 1$, the proportion remaining alive without having entered rehabilitation is $1 - \frac{10+20}{100} = 0.7$. In $t_0 + 2$, $\frac{7}{70} = 0.1$ is the share of people who enter rehabilitation, but since only 70% of the original population was alive without having entered rehabilitation at the beginning of $t_0 + 2$, the contribution to the incidence is $0.7 \times 0.1 = 0.07$, hence the cumulative rehabilitation probability is 0.17

and conducting a Wald test (Wald, 1943) to determine whether the differences across time are significant for each variable. Following divergences in trends prior to t_0 , we repeated the matching exercise with three lags, i.e. $t_0 - 3, t_0 - 2, t_0 - 1$ instead of only $t_0 - 1$. Since we require that individuals have an observed record, the lag-3 specification considers only the time period between 2014 and 2018. We also repeat the lag-1 specification for only 2014 to 2018.

3.3.2 RQ2: Mortality

We follow each individual from $t_0 + 1$ until $t_0 + 10$ or 2023, the last year of the sample, whichever occurs first and estimate a discrete-time hazard model similar to Equation 1:

$$\begin{aligned}\lambda_{i\tau}^{\text{death}} &= \Pr(Y_{i\tau} = 1 | \text{alive at beginning of follow-up year } \tau) \\ &= \Lambda_{\text{rehab}}(\alpha_\tau + \beta D_i + \mathbf{x}_i' \gamma)\end{aligned}\tag{2}$$

where $Y_{i\tau}$ is a dummy indicating that individual i dies in follow-up year $t_0 + \tau$, α_τ are time fixed effects, D_i is a dummy indicating divorce at t_0 and $\mathbf{x}_i$ are other variables measured prior to t_0 . Just as in Equation 1, we estimate two specifications for Equation 2 by varying the inclusion of variables in $\mathbf{x}_i$ based on absolute standardized mean differences after matching. We complement the pooled hazard odds ratio with cumulative mortality risks using the Kaplan-Meier estimator (Kaplan and Meier, 1958). We again report cumulative mortality risk among treated and controls, the risk difference per 1000 individuals and the corresponding risk ratio at the 5-year horizon¹⁵. We also run the lag-3 specification with more pre-divorce years.

In addition to the straightforward analysis of divorce on mortality, we run a mediation analysis to determine how much of the mortality difference can be attributed to ‘early’ musculoskeletal rehabilitation. We use landmark periods of $L \in \{1, 2\}$. This is to account for individuals who die early without a rehabilitation ever being recorded since this individual would be unmediated. We chose one or two years because a longer mediator window may have the potential to invite health shocks unrelated to divorce to produce musculoskeletal rehabilitation. Furthermore, since our data is only until 2023, a longer mediator window shortens the mortality follow-up window. With $L = 1$, the horizon, H_L is four years and with $L = 2$, it is three years. The mediator for individual i is defined as:

$$M_i(L) = \begin{cases} 1, & \text{if individual } i \text{ enters musculoskeletal rehabilitation} \\ & \text{between } t_0 + 1 \text{ to } t_0 + L, \\ 0, & \text{otherwise.} \end{cases}\tag{1}$$

We excluded matched pairs where at least one individual died before and at $t_0 + L$ or does not have an annual record in the mediator window. This is to ensure that $M_i(L) = 0$

¹⁵The hazard model uses up to ten years of follow-up although later follow-up years are available only for earlier divorce cohorts.

indicates that the individual was observed throughout the mediator window without entering rehabilitation instead of rehabilitation cannot be observed because the individual died or disappeared. We then estimate the probability of early musculoskeletal rehabilitation:

$$\Pr(M_i(L) = 1|D_i, \mathbf{z}_i) = \Lambda_{\text{rehab}}(\kappa + \delta D_i + \mathbf{z}_i' \eta) \quad (2)$$

where $\mathbf{z}_i$ contains all available pre-treatment matching variables for the relevant lag-1 and lag-3 specifications. Mortality follow-up begins after the mediator window has closed, i.e. at $t_0 + L + 1$. For the post-landmark follow-up year $t_0 + L + s$ where $s \in \{1, \dots, H_L\}$, we estimate the probability of an individual i dying given that the individual was alive at the beginning of post-landmark year $t_0 + L + s$:

$$\begin{aligned} \lambda_{is}(L) &= \Pr(Y_{is} = 1 | \text{alive at the beginning of post-landmark year } t_0 + L + s) \\ &= \Lambda_{\text{death}}(\alpha_s + \theta_D D_i + \theta_M M_i(L) + \theta_{DM} D_i M_i(L) + \mathbf{z}_i' \rho) \end{aligned} \quad (3)$$

where α_s indicates time fixed effects and D_i is a dummy indicating divorce as before. $D_i M_i(L)$ is an interaction term between divorce and early musculoskeletal rehabilitation. After predicting the probability of death in the follow-up year $t_0 + L + s$ conditional on survival to the beginning of the year, we calculate the individual-specific scenario cumulative mortality risk:

$$\hat{R}_i(d, m) = 1 - \prod_{s=1}^{H_L} [1 - \hat{\lambda}_{is}(d, m)] \quad (4)$$

where $d, m \in \{0, 1\}$ indicate divorce and rehabilitation. The scenarios are the four divorce and rehabilitation combinations: $(d, m) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. We then compute the standardized population mortality risk for the four scenarios:

$$\hat{Y}(d, M(d')) = \frac{1}{N_L} \sum_{i=1}^{N_L} [\hat{p}_i(d') \hat{R}_i(d, 1) + (1 - \hat{p}_i(d')) \hat{R}_i(d, 0)] \quad (5)$$

where N_L is the number of individual observations in the sample for landmark L and $\hat{p}_i(d') = \widehat{\Pr}(M_i(L) = 1|D_i, \mathbf{z}_i)$ is individual i 's probability of early musculoskeletal rehabilitation when divorce status is set to d' . We assume that there are no unmeasured confounding of the early rehabilitation and mortality relationship and no mediator-outcome confounders caused by divorce. We can then decompose the total effect, $Y(1, M(1)) - Y(0, M(0))$, into the natural indirect effect, $Y(1, M(1)) - Y(1, M(0))$ and the natural direct effect, $Y(1, M(0)) - Y(0, M(0))$ and interpret the decomposition as evidence about a possible rehabilitation channel.

4 Results

4.1 Descriptive summary

How many individuals are in the 20% dataset? How many person years? How many people have undergone divorce in each year? We limit matching to years 2012 to 2018 and 2014 to 2018. Who do we filter out? How many people undergo medical rehabilitation? What are the top rehabilitation codes and cases?

Of the N_T eligible treated individuals, N_M were successfully matched, corresponding to a matching rate of (X%). Because matching is 1:1 and without replacement, the matched sample contains N_M treated individuals and N_M unique controls. We additionally report the number of unmatched treated individuals, matching rates by treatment year, the distribution of propensity scores, and the mean and maximum propensity-score distances between matched individuals.

figure to show the event study coefficients and maybe Wald test - joint coefficient test

lag1 vs lag3 specification comparison with sample size and number of matches maybe?

4.2 RQ1: Musculoskeletal rehabilitation

4.3 RQ2: Mortality

5 Discussion

6 Conclusion

7 Appendix

7.1 Subsection 1